C++ in Quantitative Finance

#5

A random numbers class

Paweł Sakowski

Spring 2026

University of Warsaw
Faculty of Economic Sciences

Accuracy of results in the MC experiments

- When estimating the result using MC methods we usually calculate the average value of possibly many single results.
- The error of such approximation depends on both:
  (1) the number of simulations and
  (2) standard error of results in a single MC simulation.
- The problem of accuracy of results is strictly linked with Central Limit Theorem.

Central Limit Theorem I

- Central Limit Theorem states that if X_i are identically and independently distributed random variables with constant expected values μ and constant and finite variances σ^2 then a random variable defined as:

$$\frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}}$$

  converges in distribution to standard normal distribution for $n \to \infty$.

- In other words, we can say that the mean from n-elements sample has a normal distribution:

$$\bar{X}_n \sim N(\mu, \frac{\sigma}{\sqrt{n}})$$

Central Limit Theorem II

- As a result, we can find $(1-\alpha)$% confidence limits for the mean $\bar{X}_n$:

$$\left[\bar{X}_n - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X}_n + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

where:

  - n is the number of simulations in Monte Carlo experiment,
  - σ is standard deviation of the result in a single MC simulation,
  - $z_{\alpha/2}$ is a $(\alpha/2)$% quantile of $N(0,1)$.
  - $z_{1-\alpha/2}$ is a $(1-\alpha/2)$% quantile of $N(0,1)$.

Increasing number of simulations

- The easiest way to raise accuracy of MC results is to increase the number of simulations n.
- However, this approach is not efficient since reduction of results error is proportional to $1/\sqrt{n}$ rather than $1/n$.
- Consequently, this requires relatively high number of simulations which can result in unacceptable long duration time of the MC experiment.
- An alternative solution is to reduce standard deviation σ of the result in the single MC simulation.

Variance reduction techniques

- There are many ways to reduce variance of MC results:
  1. **antithetic variates**
  2. control variates
  3. stratified sampling
  4. moment-matching
  5. low-discrepancy sequencing, quasi-random sequences
  6. importance sampling
  7. random number re-usage across experiments
- One of the most popular is **antithetic variates** method.

Antithetic variates I

- Assume, that the result of a single MC simulation is calculated as the average from some two MC simulations. Therefore, we can write:

$$\bar{Y} = \frac{Y_1 + Y_2}{2}$$

and

$$\mathrm{Var}(\bar{Y}) = \frac{\mathrm{Var}(Y_1) + \mathrm{Var}(Y_2) + 2\mathrm{Cov}(Y_1, Y_2)}{4}$$

- If Y_1 and Y_2 are independent from each other then:

$$\mathrm{Var}(\bar{Y}) = \frac{\mathrm{Var}(Y_1)}{2} = \frac{\mathrm{Var}(Y_2)}{2}$$

Antithetic variates II

- However, $\mathrm{Var}(\bar{Y})$ can be additionally reduced if

$$\mathrm{Cov}(Y_1, Y_2) < 0$$

- The solution is to apply antithetic variates:

$$\begin{aligned} X_i \sim N(0,1) \quad &\longrightarrow \quad -X_i \sim N(0,1) \\ Z_i \sim U(0,1) \quad &\longrightarrow \quad 1 - Z_i \sim U(0,1) \end{aligned}$$

- Hence, we can easily double the sample size and additionally reduce variance of results due to to negative covariance between the results from two simultaneous simulations.

Pseudo-random numbers generators

- Let's consider a generator of pseudo-random numbers distributed as $\sim U(0,1)$. These numbers are typically generated using the following scheme:

$$y_{i+1} = (ay_i + c) \bmod m, \quad i = 0, 1, \ldots, T \tag{1}$$

$$R_{i+1} = y_{i+1}/m, \quad i = 0, 1, \ldots, T \tag{2}$$

  where a is a multiplier, c and m are constants, whereas y_0 is a **seed**, which is a place where the generator starts its work.
- Using this algorithm we can transform a given series of uniforms into series from any distribution of known cumulative distribution function (CDF).
- Since this algorithm is strictly deterministic, we call the numbers we get *pseudo*-random, rather than random.

Exercises

1. Modify Project11 to allow the user to enter the seed for the generator. If the user enters 0, then the seed should be "randomized".
2. Modify Project11 to compare for various cases convergence of Monte Carlo simulations with and without anthithetic sampling.

Thank you!

Paweł, Sakowski
sakowski@wne.uw.edu.pl